

HIGH-THROUGHPUT DISCRETE DECISION ENGINE

Zero-Token Decision Systems in Creative Production

Probabilistic classification, real-time edit pacing, and Dynamic Creative Optimization (DCO) for video editing, Figma workflows, and Adobe Creative Cloud pipelines at wire speed.

5.8 μs	0.0%	\$0.00	0 MB
INFERENCE LATENCY (P50)	ORDER PERMUTATION FLIP	PER-TOKEN API OVERHEAD	EXTERNAL WEIGHTS NEEDED

THE PARADIGM: EVALUATIVE DECISION COMPUTE VS. GENERATIVE LLMs

Creative software is not bottlenecked by asset generation; it is bottlenecked by **discrete decision latency**. Deciding which layout template, transition curve, or B-roll bin to select via cloud LLMs introduces 1,500 ms round-trips, non-deterministic hallucinations, and runaway API bills. Zev **replaces the LLM in the decision loop**, evaluating candidate options at 60 fps directly on the local machine with 100% order invariance.

01 // Video Editing & Post-Production Pipelines

1.1 Automated Cut & Transition Style

PREMIERE / RESOLVE

Evaluates dialogue cadence, pause duration, and emotion to pick timeline cut types at interactive scrubbing speeds.

INPUT CONTEXT	EVALUATED CHOICES
"Wait, what? No way... [1.4s pause]"	A: dramatic_zoom_punch
Speaker: Guest reaction	B: reaction_shot_b_roll ← [96.4%]
Goal: YouTube High-Energy	C: hold_speaker_wide
	D: jump_cut_fast

1.2 Semantic B-Roll & Media Pool Match

ASSET INGEST

Maps voiceover sentences directly into stock footage bins. Automatically abstains when no clip fits the context.

INPUT CONTEXT	EVALUATED CHOICES
"Interest rates surged and tech valuations plummeted."	A: generic_city_skyline
Available: 8 Stock Bins	B: stock_charts_red_down ← [98%]
	C: office_worker_stressed
	D: __insufficient__ (Abstain)

1.3 16:9 → 9:16 Shorts Re-Framing

MULTI-PLATFORM

Selects dynamic vertical crop anchors and split layouts per beat when repurposing landscape podcasts for TikTok/Reels.

INPUT CONTEXT	EVALUATED CHOICES
Active Faces: 2 (Simultaneous)	A: single_speaker_center
Motion: Low Audio: Cross-talk	B: split_screen_dual ← [94.2%]
Target: 9:16 TikTok Safe Zones	C: blurred_side_letterbox

1.4 Audio Ducking & Stem Hierarchy

MULTI-TRACK MIX

Routes audio stems to ducking profiles on-the-fly, balancing explosive SFX, background beds, and dialogue priority.

INPUT CONTEXT	EVALUATED CHOICES
Stem Levels: Dialogue peak -6dB	A: duck_music_24db_dialogue ←
Music Bed: Uptempo EDM -12dB	[91%]
SFX: Sub-bass drop detected	B: swell_music_emotional
	C: sfx_priority_explosive

02 // Graphic Design & Creative Automation

2.1 Dynamic Creative Optimization (DCO)

AD SERVER PIPELINE

Assembles personalized display ads in microseconds, matching discount depth, audience intent, and copy length to layout IDs.

INPUT CONTEXT	EVALUATED CHOICES
Copy: "Flash Sale: 70% Off Boots"	A: bold_sale_yellow ← [97.5%]
Length: 28 Chars (Short)	B: editorial_minimal_dark
Audience: Urgent Deal Seeker	C: multi_product_carousel

2.2 Typographic Hierarchy & Font Pairing

FIGMA / UXP PLUGIN

Adjusts font scales and pairs dynamically as designers edit text containers or translate copy across language expansions.

INPUT CONTEXT	EVALUATED CHOICES
Language: German (240 Chars)	A: condensed_grotesk_tight ← [92.8%]
Width: 320px (Mobile Sidebar)	B: wide_display_serif_hero
Brand Tone: Technical Infra	C: standard_sans_compact

2.3 Automated Export Pipeline Triage

DAM ASSET INGEST

Classifies uploaded design assets by color profile, alpha channel, and DPI into optimal raster/vector export pipelines.

INPUT CONTEXT	EVALUATED CHOICES
Channels: RGBA (Alpha transparent)	A: print_cmyk_vector_pdf
Resolution: 4096x4096px 300 DPI	B: transparent_avif_webp ← [95.1%]
Target: Web Component	C: lowres_watermarked_proof

2.4 Client Revision Scope & Severity

SLACK / PM INGEST

Classifies incoming client feedback into structured scope tiers (0 to 3) using Zev's calibrated ordinal severity ladder.

INPUT CONTEXT	SEVERITY OUTPUT
"Video is almost ready! Can we just make our logo 15% larger in lower third and swap the final URL?"	0: Cosmetic tweak (Logo/Text) ← [0]
	1: Asset replacement (B-roll)
	2: Structural re-edit (VO redo)
	3: Full reconcept / scope creep

03 // Architectural Matrix: Zev vs. Cloud LLMs

EVALUATION DIMENSION	CLOUD LLM (GPT-40 / CLAUDE)	ZEV ENGINE (ZEV-RS)	PRODUCTION IMPACT
Inference Latency	600 ms – 2,500 ms	5 – 25 microseconds (p50: 5.8 µs)	Permits 60 fps real-time timeline scrubbing & slider feedback
Operating Cost (10k ops)	\$5.00 – \$25.00 per batch	\$0.00 (Zero API or token cost)	Viable for automated high-volume rendering clusters
Offline / On-Set Usage	Fails without active internet	100% On-Device / Air-Gapped	Runs in on-set video village laptops with zero connectivity
Order Invariance	Unstable (Permuting options flips choice)	100% Invariant (0.0% permutation flip)	Guarantees deterministic, reproducible video rendering
Hallucination Risk	Invented IDs or malformed JSON syntax	0% (Strictly bounded to provided schema)	Prevents downstream automated rendering crashes
Host Integration	Cloud REST API (requires API keys)	WASM, C-ABI (.dylib/.dll), CLI, Local Daemon	Embeds natively in Adobe CEP/UXP, Premiere, and Figma plugins

DEPLOYMENT RECOMMENDATIONS FOR CREATIVE SOFTWARE

• Adobe Premiere Pro & After Effects (Desktop): Link against Zev via C-ABI dynamic library (`.dylib` / `.dll`) or run `zev serve` as a lightweight localhost daemon (`http://127.0.0.1:8080`).

• Figma & Webflow (Web / Canvas): Compile `zev-rs` to WebAssembly (`wasm32-unknown-unknown`) to run 100% in-browser without sending user creative data across the wire.